

Reyhaneh Hosseini

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Education

PhD	EPFL , EDRS (Robotics, Control, and Intelligent Systems) <i>Thesis: 3D Human Motion: From Multimodal Forecasting to Physics-Informed Generation</i> Advisor: Prof. Alexandre Alahi	2022 – present
MS	University of Tehran , Mechanical Engineering, CGPA: 19.02 / 20.0 (4.0 / 4.0) <i>Thesis: Controlling a Biped Robot with Reinforcement Learning</i> Advisors: Prof. Aghil Yousefi-koma and Prof. Massoud Shariatpanahi	2019 – 2022
BS	University of Tehran , Mechanical Engineering, CGPA: 16.89 / 20.0 (3.6 / 4.0) + Minor in Computer Engineering	2014 – 2019

Selected Publications

Physics Derived Plausibility Metrics for 3D Human-Human Motion Generation <i>Reyhaneh Hosseini</i> , T. Van Der Meer, M. Shukla, A. Alahi	Under Preparation
MotionMap : Representing Multimodality in Human Pose Forecasting <i>Reyhaneh Hosseini</i> *, M. Shukla *, S. Saadatnejad, M. Salzmann, A. Alahi	CVPR 2025
HHI-Assist : A Dataset and Benchmark of Human-Human Interaction in Physical Assistance S. Saadatnejad *, <i>Reyhaneh Hosseini</i> *, J. Barreiros, K. Tsui, A. Alahi	RA-L 2025
Posedriver : A Unified Approach to Multi-Category Skeleton Detection for Autonomous Driving Assistance Y. Borhani, T. Mordan, Y. Wang, <i>Reyhaneh Hosseini</i> , J. Khoramdel, A. Alahi	Under Review

Experience

Visual Intelligence for Transportation Lab (VITA) Doctoral Assistant EPFL, Switzerland	2022 – present
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- Ongoing: Developing *physics-informed motion generation* for human interactions. The simulations were built using the *Newton* physics engine (*NVIDIA Warp*).
- Ongoing: Collaborating with the NeuroRestore lab to develop a virtual patient and stimulation-to-EMG modeling framework for spinal cord injury patients.
- Achieved state-of-the-Art results in *human motion forecasting* by modeling **multimodality** and **uncertainty**. We achieved this by representing multimodality through a novel heatmap-based representation, also leading to better language alignment.
- Developed the 'HHI-Assist' **dataset** to benchmark the advances in assistive robotics research. This dataset addresses the scarcity of data on close-contact human-human interaction.
- Developed a novel data-creation pipeline to build a large-scale Persian dataset. Finetuned an LLM-based **Text-to-Speech (TTS)** model on this data using both full-parameter and LoRA.
- Utilized **cross-task consistency** to improve 3D human pose estimation. This framework measured coherence between direct 3D estimation and 2D to 3D pose lifting.
- Applied the TrajNet++ framework for human trajectory forecasting to predict the motion of professional hockey players from positional data.
- **Mentorship**: Guided multiple Master's students through the following semester projects:
 - Adapting the HHI-Assist dataset for *SMPL-compatible 3D human modeling*.
 - Implementing 3D car pose estimation from monocular images.
 - Applying the OpenPifPaf framework for 2D/3D lane detection and pose estimation.

Center for Advanced Systems and Technologies (CAST)

2017 – 2022

Research Assistant | *University of Tehran, Iran*

- Designed a **Reinforcement Learning-based controller** for the Surena IV humanoid's locomotion, utilizing **PyBullet** for simulation and validation (Master's Thesis).
- Engineered and built a smart-actuated **prosthetic hand**, designing its embedded circuitry and implementing an intuitive EMG-based control system. (Undergraduate Final Project)
- Developed walking pattern simulations for the **Surena IV robot** in the **Choreonoid dynamics simulator** (Undergraduate Research).

Applied Artificial Intelligence Lab

2020 – 2022

Research Assistant | *University of Tehran, Iran*

- Initiated a framework for detecting Keratoconus from corneal topography images.

Research Center for Biomedical Technologies and Robotics (RCBTR)

2016

Research Intern | IK Hospital Complex, Tehran, Iran

- Designed calibration phantoms for micro CT-scans.
- Designed a 3D printed mold and fabricated silicone-rubber breast tumors for biopsy training. Developed training tools of the Sina surgical robot.

Teaching and Supervision

Teaching Assistant

EPFL

Deep Learning for Autonomous Vehicles; Introduction to Machine Learning for Engineers

2022 – 2026

Teaching Assistant

University of Tehran

Artificial Intelligence; Measurement Systems and Instrumentations;
System Identification; Introduction to Mechanical Engineering

2017 – 2021

Technical Skills

Programming: Python, C++, C

Machine Learning & Vision: PyTorch, TensorFlow, Scikit-learn, Hugging Face, SMPL/SMPL-X

Simulation & Robotics: NVIDIA Warp, Newton, PyBullet, Choreonoid

DevOps & Tools: Linux, Git, Docker, MATLAB, SolidWorks

Honors

Graduate Studies Fellowship, University of Tehran (*Offered merit-based MS admission*)

2019

2nd Place, World Robot Olympiad (WRO), Iran

2016

Top 0.18% National Ranking (397^{th} / $222,507$ in *University Entrance Exam*)

2014

Professional Service

Reviewer, IEEE/CVF International Conference on Computer Vision (ICCV)

2025

Volunteer Experience

Member, IRSA (Iranian Students Association at EPFL)

2022 – 2023

SAME (Scientific Association of Mechanical Engineering)

2016 – 2017

- Member of the committee for the Mechanical Engineering freshmen welcome ceremony.

Languages

Persian (Native) **English** (Fluent) **French** (A2)

Personal Interests

Clay Art, Puzzling, Skiing, Skating